Multiple Recovery Paths in a Lesioned Transformer

Internal Rerouting and Readout Remapping

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Motivation

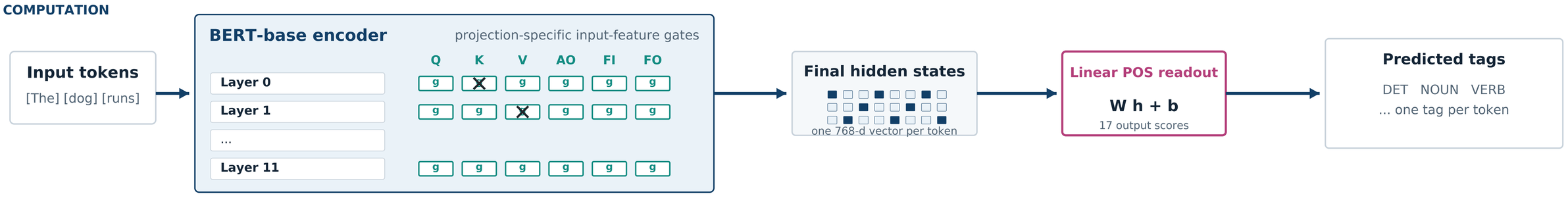
Using BERT as a frozen encoder and a linear part-of-speech (POS) readout as a task classifier, we first use gate masking to learn a sparse core subnetwork sufficient for POS classification. We then ablate this subnetwork to produce a large drop in performance. Our main question is whether performance can recover by adapting the surviving projection-specific gates, retraining the linear POS readout, or both.

CENTRAL FINDING

Recovery of performance does not require recovery of the original representation.

Experimental framework

COMPUTATION



EXPERIMENTAL PROGRESSION

A Learn the task-sufficient core

Train projection-specific hard-concrete gates and the linear POS readout while pretrained encoder weights remain fixed.

Gates	TRAIN
Readout	TRAIN
Encoder weights	FIXED

The sparse configuration reaches 94.98% primary-run POS accuracy.

B Apply a permanent lesion

Mask scores forced to -12; effective gates are zero and gradients are blocked.

74 Q/K/V 64-coordinate input groups
5 attention-output gate tensors
9 FFN-intermediate + 4 FFN-output tensors

No recovery pathway can reactivate these entries.

C Recovery trainability

Condition	Gates	Readout	Encoder	Lesion
Gates only	TRAIN	FIXED	FIXED	OFF
Readout only	FIXED	TRAIN	FIXED	OFF
Gates + readout	TRAIN	TRAIN	FIXED	OFF
Encoder only	FIXED	FIXED	TRAIN	OFF

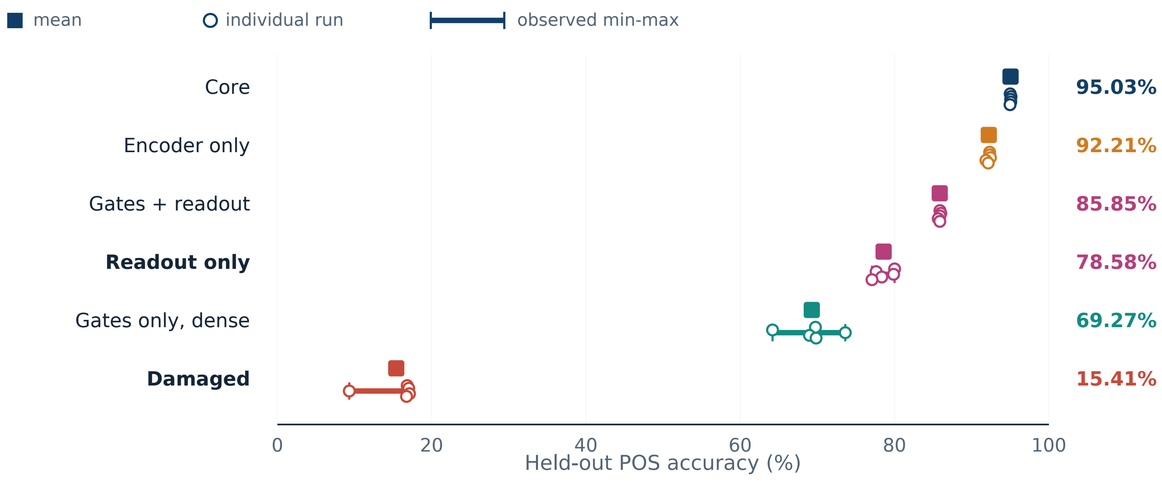
SETUP bert-base-uncased | UD English-EWT | 17 POS tags | 25,147 development tokens

LESION 74 Q/K/V groups + 18 dense projection-mask tensors; 37,248 gate entries clamped

Schematic only: arrows show computation or experimental progression, not measured activation flow or traced causal paths.

Recovery performance

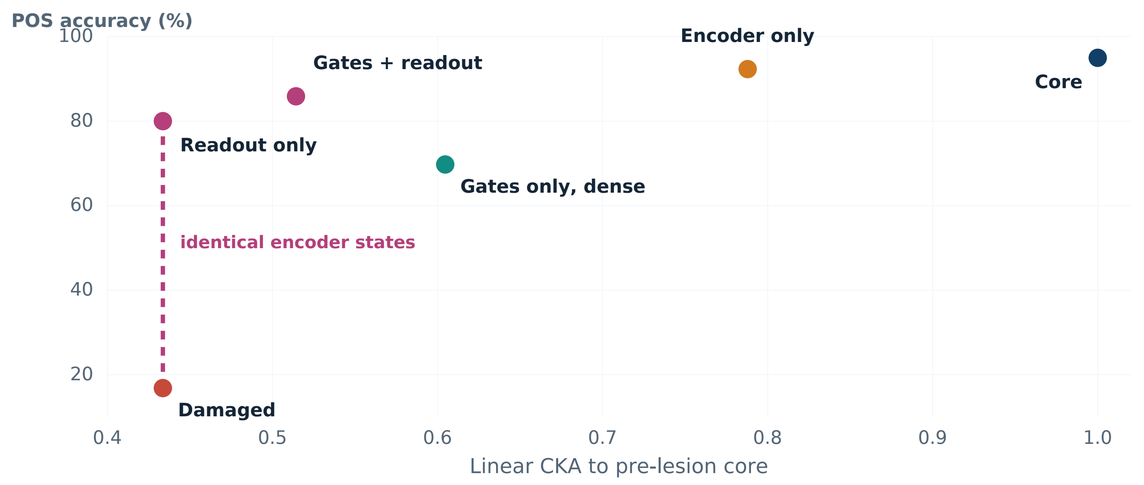
Mean across five seeds



All recovery conditions produced marked improvement over the damaged baseline. Filled squares show arithmetic means across five full-pipeline runs. Open circles are individual runs; capped spans show observed min-max ranges, not confidence intervals. The gate-only result is the density-promoting variant.

Accuracy versus representation

Primary run

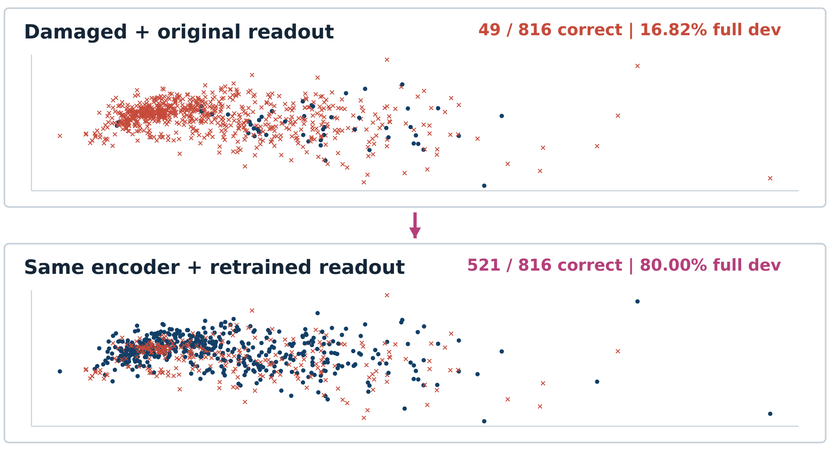


Representational analysis showed that behavioral recovery was not necessarily accompanied by restoration of the original hidden-state geometry. CKA measures hidden-state similarity, not recovered circuitry. Readout-only leaves encoder states unchanged yet raises primary-run accuracy from 16.82% to 80.00%.

Different recovery constraints produce different representational geometries

Readout-only recovery

Same token representations; only the linear readout changes

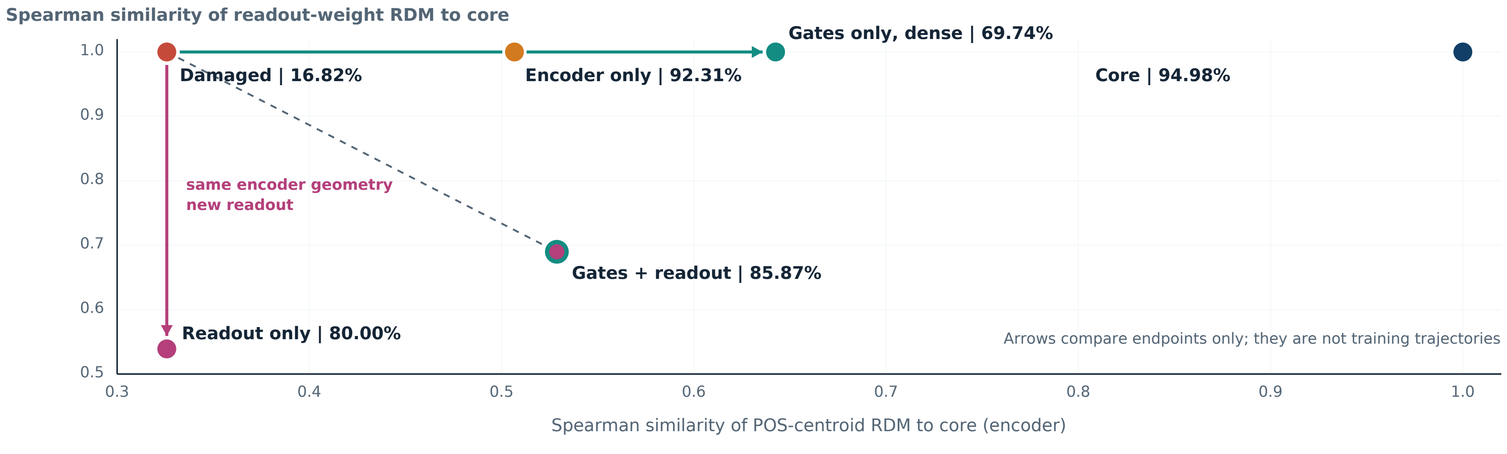


Coordinates are fixed across panels; only correctness under the original versus retrained readout changes. The shared core-fitted PC1-PC2 view explains 23.32% of core variance, while classification uses all 768 dimensions.

Encoder and readout geometry

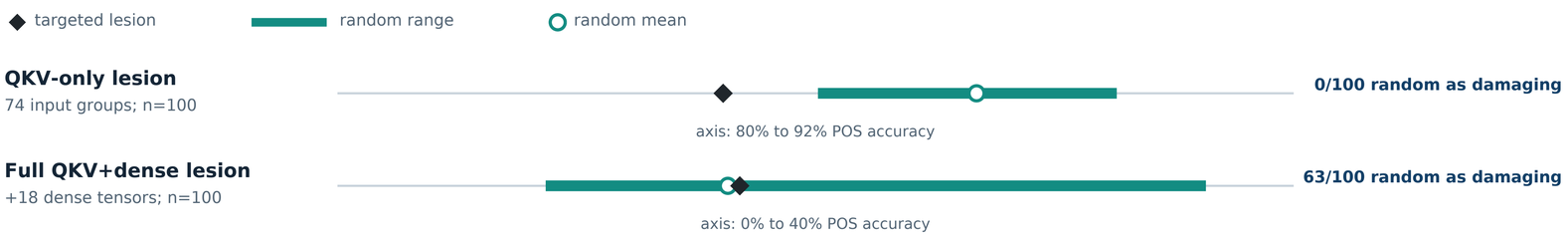
Primary-run endpoints; labels report held-out POS accuracy

Spearman similarity of readout-weight RDM to core



Readout-only holds encoder POS-centroid similarity at 0.33 while readout-weight similarity changes from 1.00 to 0.54 and accuracy reaches 80.0%. Gate-only and joint recovery also move the encoder geometry.

Robustness and controls



The QKV-only control supports enrichment relative to globally random QKV groups. The full random QKV+dense control does not establish unique specificity of the exact dense target set; it qualifies the causal interpretation of the severe lesion.

Conclusions

Performance can recover without relearning the underlying transformations.

The extent to which the original geometry is recovered depends on the degrees of freedom of the recovery protocol.

Circuit necessity is conditional on the permitted rerouting strategy. A component may be necessary under the original fixed routing and readout, but cease to be so when the readout can be retrained or surviving transformations can be reselected.

Our evaluations were based on a single task and a single backbone model. The biological relevance of these findings remains unclear.

Methods: Louizos et al. (2018); Cao et al. (2021). Data: Universal Dependencies English-EWT.